

6(a)(i)	$P = VI$	C1
	$I = 30 / 120$ $= 0.25 \text{ A}$	A1
6(a)(ii)	$Q = 0.25 \times 3.0 \times 3600 (= 2700)$	C1
	number = $(0.25 \times 3.0 \times 3600) / 1.60 \times 10^{-19}$ $= 1.7 \times 10^{22}$	A1
6(b)	$R = V/I$ or $R = P/I^2$ or $R = V^2/P$	C1
	$= 120 / 0.25$ or $= 30 / 0.25^2$ or $= 120^2 / 30$ $= 480 \Omega$	A1

Question	Answer	Marks
6(c)	$R = \rho l / A$	C1
	$A = (6.1 \times 10^{-7} \times 580 \times 10^{-3}) / 480 (= 7.37 \times 10^{-10})$	C1
	$d = [(4 \times 7.37 \times 10^{-10}) / \pi]^{1/2}$ $= 3.1 \times 10^{-5} \text{ m}$	A1
6(d)	temperature decreases and so resistance decreases	B1

Question	Answer	Marks
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